



San Francisco's Hidden Overdose Season

This presentation uncovers the seasonal patterns in San Francisco's opioid overdose crisis, drawing on medical examiner data from 2020–2025. By revealing when risks peak, it empowers public health officials, harm reduction managers, and policy planners to deploy timely interventions, saving lives through proactive strategies.

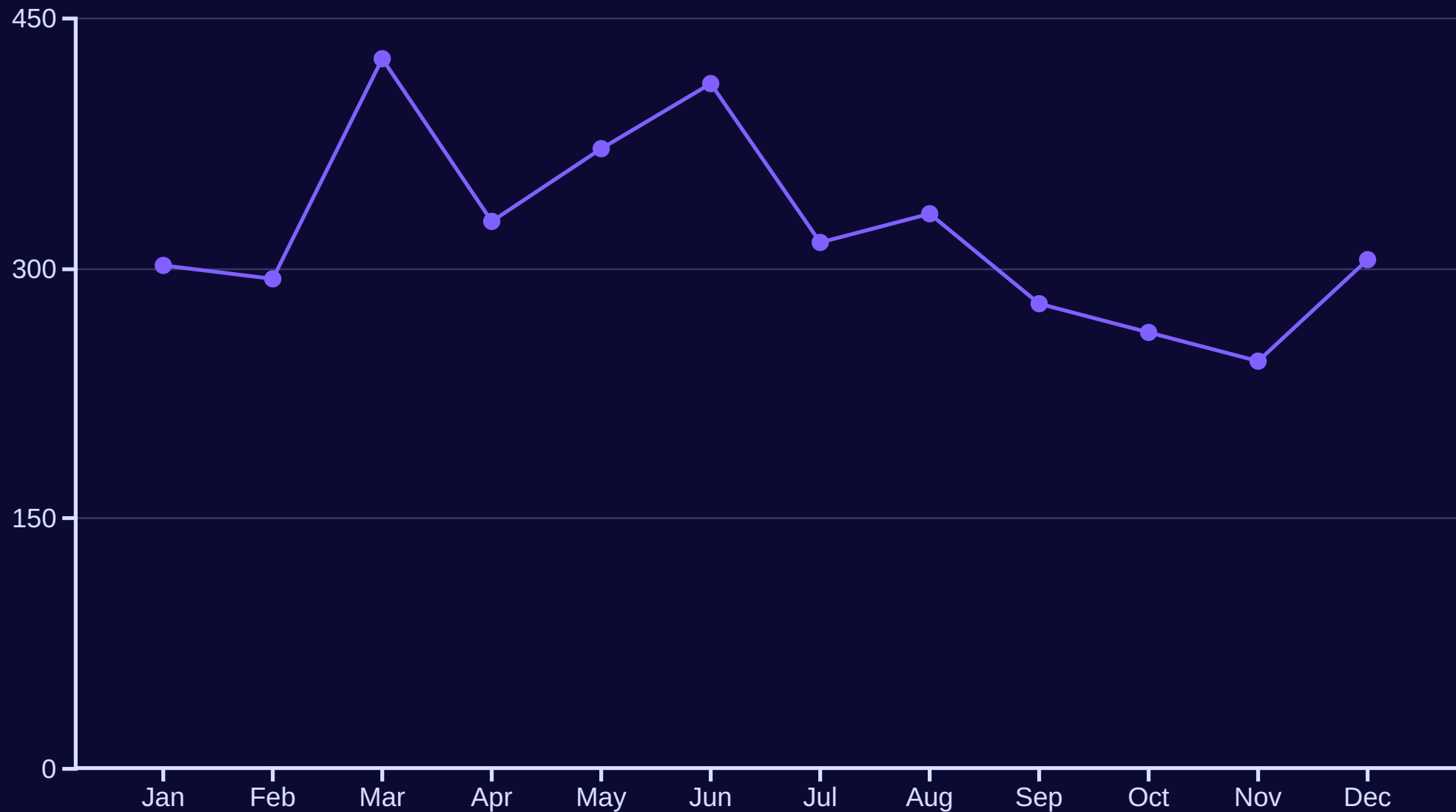
San Francisco's Overdose Crisis: Understanding When Risk Peaks

The city has faced a persistent **opioid overdose emergency**, claiming hundreds of lives each year. Using data from the **San Francisco Office of the Chief Medical Examiner (2020–2025)**, this project explores **when** overdose deaths are most likely to occur. Understanding these timing patterns helps public health teams **plan interventions proactively** rather than reactively. Evidence shows that seasonal surges strain resources, but foreknowledge allows for compassionate, evidence-based responses to mitigate harm and support vulnerable communities in San Francisco.

Data source: Preliminary Unintentional Drug Overdose Deaths, 2020–2025.



San Francisco's Hidden Overdose Season



Overdose deaths follow a **seasonal rhythm** — rising sharply each **spring and early summer (March–June)**. Monthly deaths average around **323**, but peak at **426 in March** and dip to **244 in November**. Nearly **one in four months exceeds 350 deaths**, marking recurring high-risk periods. This urgent pattern demands immediate attention to protect lives compassionately.

Avg = 323 | Peak = 426 | 25% exceed 350

Seasonal Trends = Seasonal Strain



Emergency Room Spikes

During spring surges, emergency rooms and outreach programs experience **spikes in demand**, overwhelming staff and resources in an already strained system.



Life-Saving Interventions

Timely interventions — **naloxone distribution, treatment outreach, and awareness campaigns** — can prevent fatalities with evidence-based compassion.

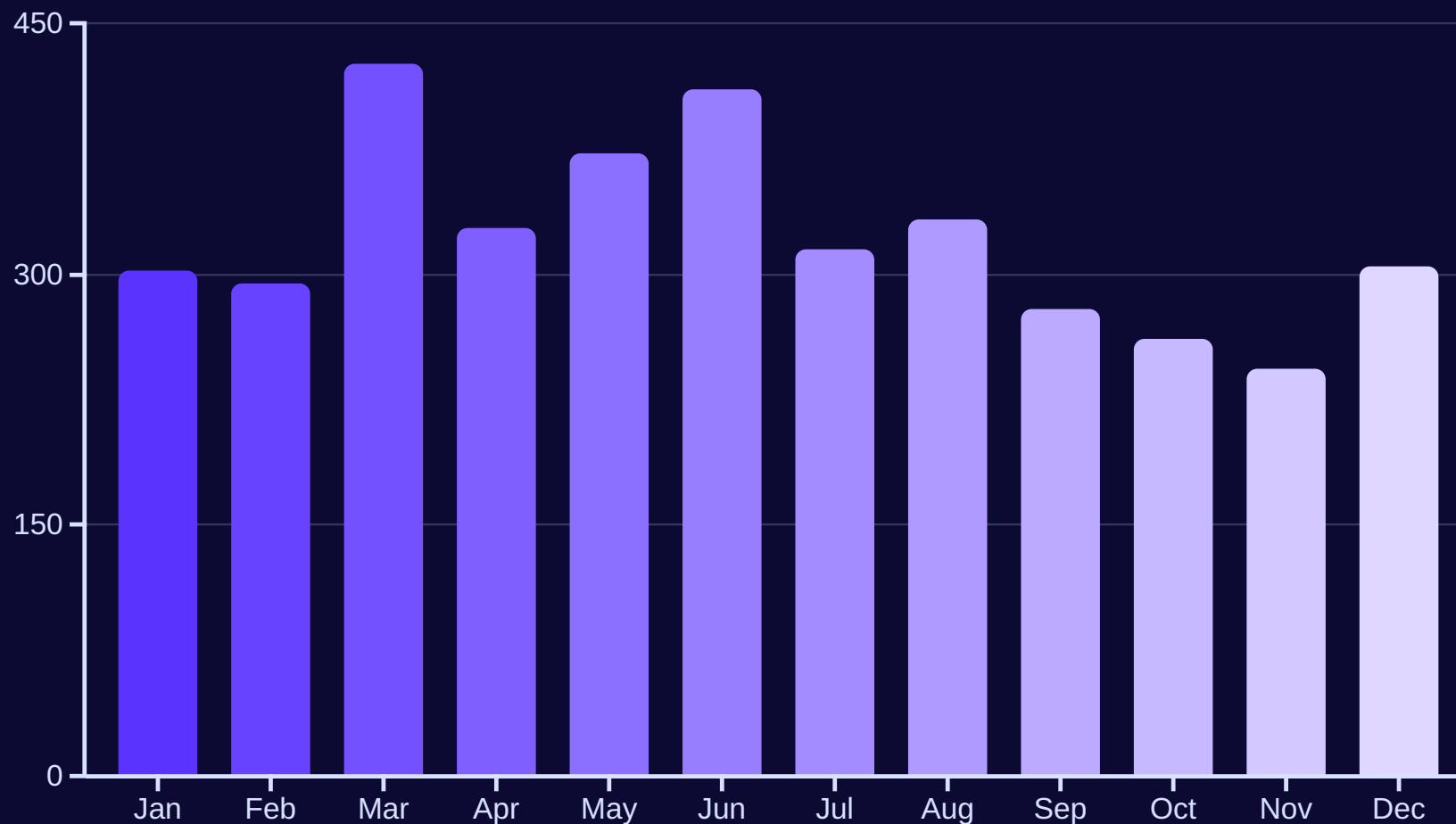


Efficient Resource Allocation

These findings matter **socially and medically**: they help public agencies save lives and allocate limited resources more efficiently across San Francisco.

“More than 400 people lost their lives to overdoses in San Francisco last March.”

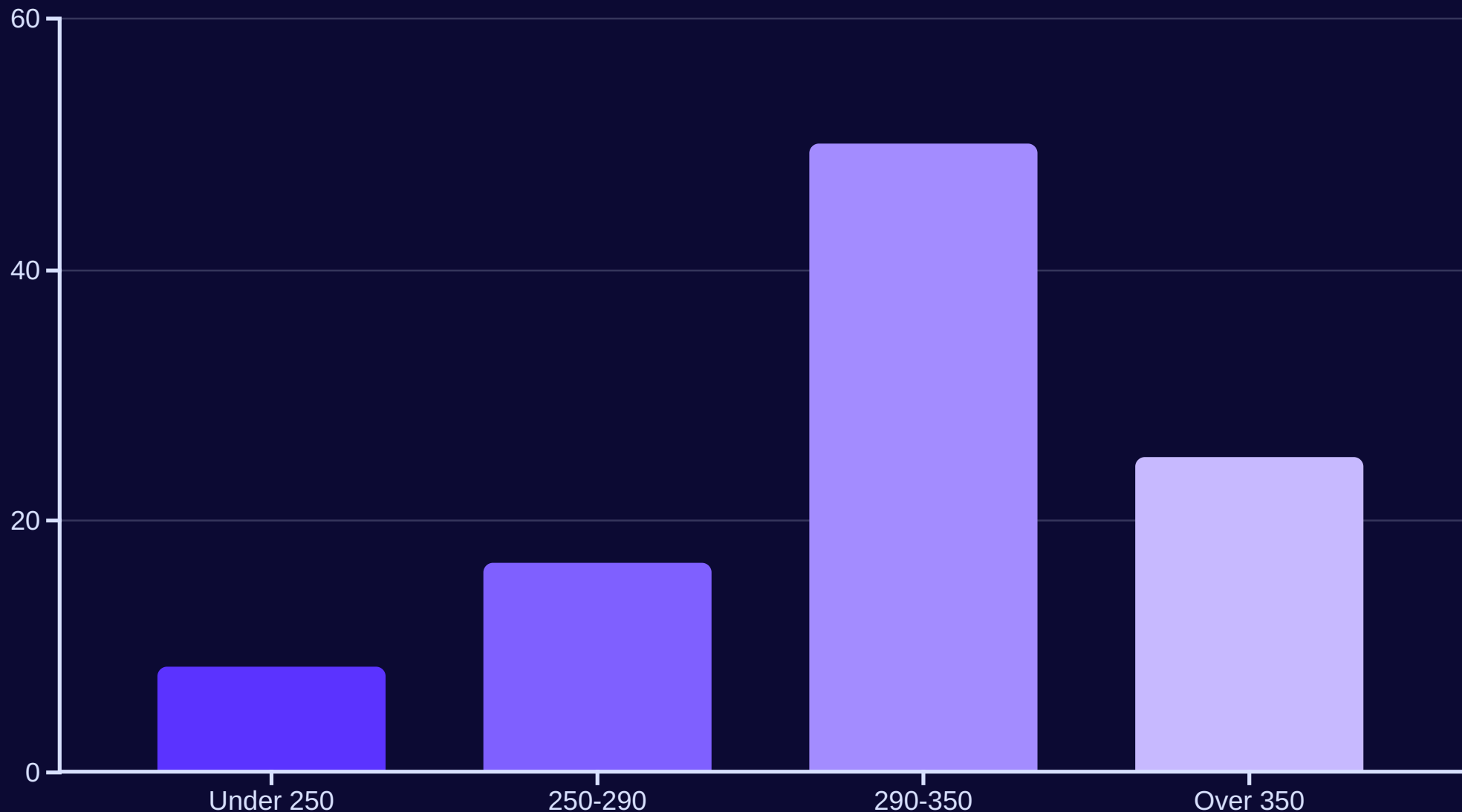
The Shape of the Year: When Deaths Rise and Fall



Aggregating five years of data reveals a consistent cycle: **Highs:** March (426), June (411). **Lows:** October (261), November (244). The **200-death gap** between these months **illustrates** how uneven the risk is across the calendar year. This evidence underscores the urgent need for seasonal planning to address peaks compassionately.

But how often do these high months occur?

High-Risk Months Are Common



Most months cluster around **290–350 deaths**, but the pattern is **skewed toward higher counts**. Around **25% of months exceed 350 deaths**, meaning “bad months” are not rare. Statistically, monthly deaths are **similar to the average**, but with **more high spikes than expected** — confirming a recurring risk pattern that demands urgent, evidence-based action.

Each bar represents aggregated monthly totals, 2020–2025.

Turning Insight Into Action

February–March: Prepare

Expand outreach, naloxone access, and resource planning to build capacity compassionately before the surge.



July–August: Monitor

Adjust strategies based on real-time data to sustain harm reduction efforts effectively.



March–June: Respond

Deploy mobile health units and overdose prevention teams to address peaks with urgent interventions.



September–November: Evaluate

Assess outcomes and plan for the next cycle, ensuring continuous improvement.



This cycle provides a roadmap for proactive public health responses, potentially averting hundreds of tragedies annually.

Mapping What Comes Next

Next Steps

- Use **neighborhood-level data** to pinpoint where the spring surge hits hardest, targeting resources efficiently.
- Integrate **demographic variables** to identify vulnerable populations for tailored, compassionate support.
- Explore **policy or environmental factors** influencing seasonality to inform evidence-based reforms.

Impact Statement

“If we act before March, hundreds of lives could be saved each year.”

This work urges immediate, proactive planning to transform seasonal insights into life-saving action across San Francisco.